Roll No							

End Semester Examination 2024

Name of the Course: Diploma CS

Semester: 4

Name of the Paper: Computer Architecture and Maintenance

Paper Code: DTCS 401

Time: 3 Hour's

Maximum Marks: 100

Note:

(i) All Questions are compulsory.
(ii) Answer any two sub-questions among a,b, and c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	**(10 X2 = 20 Marks)**	CO1 & CO2
(a)	Explain the block diagram of North Bridge / South Bridge Architecture. Also, differentiate between North Bridge and South Bridge Architecture (CO1)	
(b)	What are Servo Techniques? Explain different types of Servo Techniques (CO2)	
(c)	Compare: (a) PCI with PCI Express Bus. (b) Intel Processor P3 and P4. (CO2)	
Q2	**(10 X2 = 20 Marks)**	CO1 & CO3
(a)	What is a CRT Monitor? Explain with the help of a block diagram. Also, discuss the characteristics of CRT Monitor. (CO3)	
(b)	What is Cache memory? Differentiate Internal and External Cache in terms of L1, L2, and L3 Cache. (CO1)	
(c)	What are the components of Motherboard? State any ten motherboard selection criteria. (CO1)	
Q3	**(10 X2 = 20 Marks)**	CO2 & CO5
(a)	Explain the workings of UPS along with its types. (CO5)	
(b)	Explain the block diagram of CD-ROM. Compare CD and DVD. (CO2)	
(c)	What is Formatting? Explain Low Level and High formatting. (CO2)	
Q4	**(10 X2 = 20 Marks)**	CO3 & CO4
(a)	What is a Scanner? Explain different types of Scanners. (CO3)	
(b)	What is Switching? Explain Circuit Switching and Packet Switching. (CO4)	
(c)	Write a short note on: (a) TDM (b) OCR (c) TWAIN (d) FHSS (e) DSSS (CO4)	
Q5	**(10 X2 = 20 Marks)**	CO4 & CO5
(a)	Explain the block diagram and working of SMPS. (CO5)	
(b)	Differentiate between: (a) Dot matrix Printer vs Laser printer (b) Active-matrix vs Passive Matrix (CO4)	
(c)	What is a Keyboard? Explain the different types of keyboards that exist. (CO4)	